

Prepared By	08-Aug-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	08-Aug-2012	Mr.Chatchawan S.		
Approved By	08-Aug-2012	Mr.Amnauy W.		
Certified By	08-Aug-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	08-Aug-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-017-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-004**



RENEW COILS ENGINEER & SUPPLY CO., LTD.

WELDING PROCEDURE SPECIFICATION (WPS)

WELDING PROCEDURE SPECIFICATION (WPS) Yes ☒
PREQUALIFIED ☒ QUALIFIED BY TESTING ☐
or PROCEDURE QUALIFICATION RECORD (PQR) ☐

Company Name Renew coils and Engineering & Supply co., ltd.
Welding Process (es) SMAW
Supporting PQR No.(s) RNC-PQR-004

Identification # RNC-WPS-004
Revision 0 Date 8-Aug-12 By Mr. Chatchawan s.
Authorized by Mr. Amnuy V. Date 8-Aug-12
Type-Manual ☒ Semi-Automatic ☐
Machine ☐ Automatic ☐

JOINT DESIGN USED

Type : Butt Weld
Single ☒ Double Weld ☐
Backing : Yes ☐ No ☒
Backing Material : NONE
Root Opening 3 Root Face Dimension 3
Groove Angle : 60⁰⁻⁵ Radius (J-U) NONE
Back Gouging : Yes ☒ No ☐ Method

BASE METALS

material Spec. SS400 / A36 or Strength Equivalence
Type or (Grade) Group 1 Carbon
Thickness : Groove 3.0 mm - Unlimited Fillet N/A
Diameter (Pipe) -

FILLER METALS

AWS Specification A 5.1
AWS Classification E7016
Brand Name KOBELCO LB 52

SHIELDING

Flux N/A Gas N/A
Composition
Electrode-Flux (Class) N/A Flow Rate
Gas Cup Size

PREHEAT

Preheat Temp., Min 30 C°
Interpass Temp., Min 100C° Max 250 C°

POSITION

Position of Groove : 4G Fillet : None
Vertical Progression : Up ☒ Down ☐

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW) Short-Circuiting ☐
Globular ☐ Spray ☐
Current : AC ☒ DCEP ☐ DCEN ☐ Pulsed ☐
Other
Tungsten Electrode (GTAW)
Size : NONE
Type : NONE

TECHNIQUE

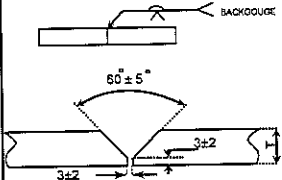
Stringer or Weave Bead : Stringer and Weave Bead
Multi-pass or Single Pass (per side) Multi-pass
Number of Electrodes
Electrode Spacing Longitudinal N/A
Lateral N/A
Angle N/A

Contact Tube to Work Distance
Peening N/A
Interpass Cleaning : Grinding or Wire Brush

POSTWELD HEAT TREATMENT

Temp. N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Current		Volts	Travel Speed cm/min	Joint Details
		Class	Diam.	Type & Polarity	(Amps) or Wire Feed Speed			
Root pass	SMAW	E7016	3.2	AC/EP	80-120	20-25	5-12	
Hot pass	SMAW	E7016	3.2	AC/EP	80-120	20-25	5-12	
Filling pass	SMAW	E7016	4	AC/EP	130-180	25-28	7-15	
cap	SMAW	E7016	3.2	AC/EP	80-120	20-25	5-12	

PREPARED BY

REVIEWED BY

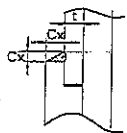
APPROVED BY :

DATE :

DATE :

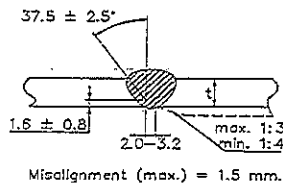
DATE : 20-8-12

ATTACHED FIGURE JOINT SKETCH



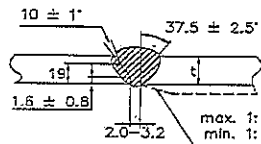
Cx (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

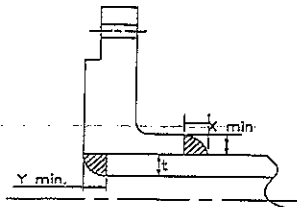
JOINT No. B1 & B2



Approx. 1.6 mm.

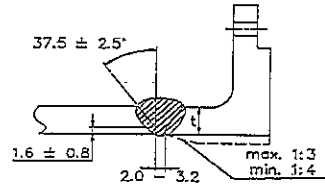
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



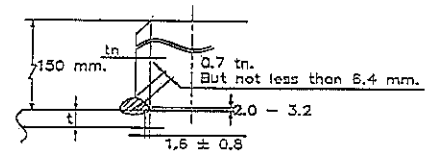
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



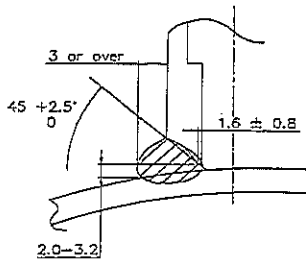
Misalignment (max.) = 1.5 mm.

JOINT No. B3

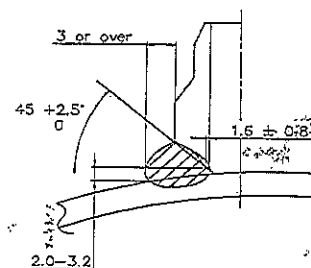


Misalignment (max.) = The lesser of 3.2 mm. or $0.5t$.

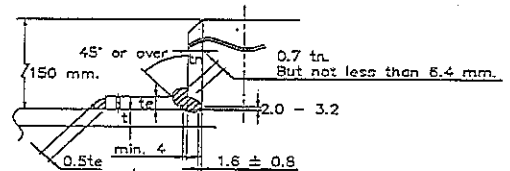
JOINT No. B4



JOINT No. C1

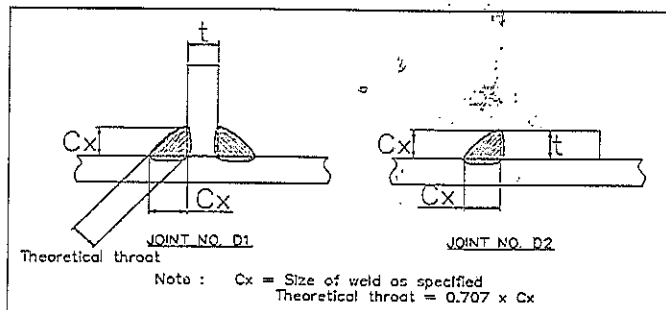


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5t$.

JOINT No. B5





PAE TECHNICAL SERVICE COMPANY LIMITED
69 Soi On-nuch 64, Srinakarin Road, Suanluang, Suanluang, Bangkok 10250
Tel. : (662) 322-0222 (Auto 20 lines) Fax. : (662) 721-2511

PQR No. : PQR-AWS-SMAW-004

PROCEDURE QUALIFICATION RECORD (PQR)

NDT and Mechanical Test Result

TENSILE TEST (Report No : TS-12-08-082)						
Specimen No.	Width mm.	Thickness mm.	Area	Ultimate Load (KN)	Ultimate Strength (N/mm. ²)	Character of Failure and Location
3181/12 TR1	20.07	25.00	501.750	216.530	431.550	OUT OF WELD
3181/12 TR2	20.41	24.84	506.984	221.041	435.992	OUT OF WELD
GUIDE BEND TEST (Report No : BD-12-08-021)						
Specimen No	Type of Bend		Test Result		Location of Indication	
3181/12 BS1	Transverse Side Bend		No Open Discontinuity		N/A	
3181/12 BS2	Transverse Side Bend		Open Discontinuity 1.16,0.19,0.43 mm		WELD ZONE	
3181/12 BS3	Transverse Side Bend		Open Discontinuity 0.85 mm		WELD ZONE	
3181/12 BS4	Transverse Side Bend		No Open Discontinuity		N/A	

Visual Inspection, PQR Report No:

PQR-013

Radiographic Examination

Appearance: Acceptable
Undercut/Slags: No Indication
Surface Porosity: No Indication
Test Date: 28 Jul 12
Witness By: Mr.Wichit K.
Organization: PAE Technical Service Co.,Ltd.

RT Report No: RT-PQR-016 Result: Accepted
Organization: PAE Technical Service Co.,Ltd.

Other Tests

1) Vicker Hardness Test

Report No: HN-12-08-034

3) Fillet Weld Test

Report No: N/A.

2) Macrostructure Examination

Report No: MA-12-08-025

4) Impact Test

Report No: N/A

Welder's Name : Mr.Choochat M.

Welder No : -

Mechanical Test Conducted By : LPN Plate Mill Public Company Limited

Test Conducted By : Mr.Keerati Sa-ngoen

Report No : -

We, the undersigned, certify the statements in this record are correct and that the test welds were prepared, welded, and Tested in conformance with the requirements of Clause Section 4 of AWS D1.1 Structural Welding Code-Steel,

Manufacturer by
State Construction Co.,Ltd.

Contractor by
PAE Technical Service Co.,Ltd.

Signature: _____
Name: _____
Date: _____

Signature: _____
Name: Mr. Chaloemkiet R.
Date: 17-Aug-2012

WELDING PROCEDURE SPECIFICATION (WPS) Yes

PREQUALIFIED

or PROCEDURE QUALIFICATION RECORD (PQR) Yes

(ACCORDANCE WITH THE REQUIREMENT OF SECT 4 AWS D1.1 2010)

Identification #

Revision 0

Authorized by

Type

#

Date

Mr.Arnuy V.

Manual

RNC-WPS-004

By Mr.Chatchawan S.

Date

Semi-Automatic

Company Name

Welding Process(es)

Supporting PQR No.(s)

Renew Coils Engineering & Supply Co.,Ltd.

SHIELD METAL ARC WELDING

RNC-PQR-004

JOINT DESIGN USED

Type :

Backling :

Backling Material :

Root Opening :

Groove Angle :

Back Gouging

Method :

Single V-Groove Butt Weld

Single

Double weld

Yes

None

3.0 mm.

Root Face Dimension :

60°±5

Radius (J-U)

Yes

No

Grinding

BASE METALS

Material Spec.

Type or Grade

Thickness :

Diameter (Pipe)

ASTM,JIS G3101

SS 400

Groove

25 mm.

Fillet :

N/A

FILLER METALS

AWS Specification

AWS Classification

A 5.1

LB-52/E7016

SHIELDING

Flux

Electrode-Flux(Class)

Gas

Composition

Flow Rate

Gas Cup Size

PREHEAT

Preheat Temp.:

Interpass Temp., Min

Ambient

38°C

Max

241°C

POSITION

Position of Groove

Vertical Progression

4G

Fillet

OH

Up

Down

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW)

Current

Other

Tungsten Electrode (GTAW)

Size

Type

Short-Circuiting

Globular

DCEN

Spray/Pulse

Pulsed

AC

DCEP

None

None

TECHNIQUE

Stringer or Weave Bead :

Multi-pass or Single Pass (per side)

Number of Electrodes

Electrode Spacing

Stringer

Multipass

Single

Longitudinal

Lateral

Angle

Contact End to Work Distance

Peening

Interpass Cleaning :

Arc stable

None

Slags remove by Grinding / Wire Brush

POSTWELD HEAT TREATMENT

Temperature

Time

N/A

N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Current		Volts	Travel Speed (cm / min)	Heat Input (kj/cm)
		Class	Dia. (mm.)	Type & Polarity	Amperage or Wire speed			
1	SMAW	E7016	3.2	DCEP	103	23	6.02	22.93
2	SMAW	E7016	3.2	DCEP	123	26	6.17	31/10
3	SMAW	E7016	3.2	DCEP	126	26	3.67	53.56
4	SMAW	E7016	3.2	DCEP	122	25	7.30	25.07
5	SMAW	E7016	3.2	DCEP	124	26	5.20	37.20
6	SMAW	E7016	3.2	DCEP	124	26	8.41	23.01
7	SMAW	E7016	3.2	DCEP	125	26	14.68	13.29
8	SMAW	E7016	3.2	DCEP	123	26	12.38	15.50
9	SMAW	E7016	3.2	DCEP	118	24	12.03	14.13
10	SMAW	E7016	3.2	DCEP	118	24	11.66	14.58
Back Weld	SMAW	E7016	3.2	DCEP	119	24	11.52	14.88
12	SMAW	E7016	3.2	DCEP	118	24	15.08	11.27

Joint Details



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

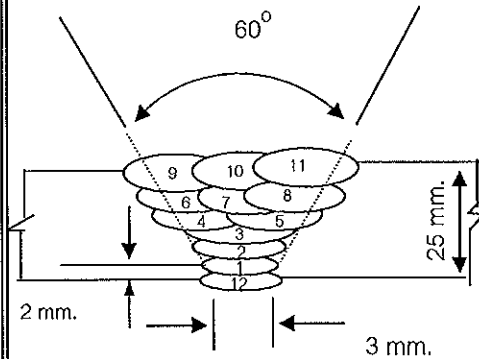
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 2

Client : Renew Colls Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-013	
Customer Order No. : N/A		WPS No. : RNC-WPS-004		Date : 6-Aug-12			

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOOCHAT M.	SMAW	4G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	SMAW	E 7016	3.2	DCEP	103	23	6.02	38	22.93	-	-	-	-
2	SMAW	E 7016	3.2	DCEP	123	26	6.17	110	31.10	-	-	-	-
3	SMAW	E 7016	3.2	DCEP	126	26	3.67	199	53.56	-	-	-	-
4	SMAW	E 7016	3.2	DCEP	122	25	7.30	223	25.07	-	-	-	-
5	SMAW	E 7016	3.2	DCEP	124	26	5.20	229	37.20	-	-	-	-
6	SMAW	E 7016	3.2	DCEP	124	26	8.41	241	23.01	-	-	-	-
7	SMAW	E 7016	3.2	DCEP	125	26	14.68	239	13.29	-	-	-	-
8	SMAW	E 7016	3.2	DCEP	123	26	12.38	228	15.50	-	-	-	-
9	SMAW	E 7016	3.2	DCEP	118	24	12.03	166	14.13	-	-	-	-
10	SMAW	E 7016	3.2	DCEP	118	24	11.66	192	14.58	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit

7-Aug-2012



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 2

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR			Location : RNC Work Shop			Report No. PQR-013	
		Customer Order No. : N/A			WPS No. : RNC-WPS-004			Date : 6-Aug-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOOCHAT M.	SMAW	4G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	SMAW	E 7016	3.2	DCEP	119	24	11.52	198	14.88	-	-	-	-
12	SMAW	E 7016	3.2	DCEP	118	24	15.08	215	11.27	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit

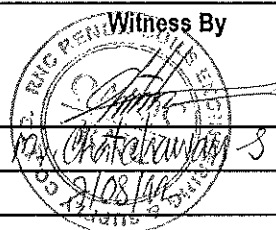
7-Aug-2012



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjatee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th



23854

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-08-082
Revision No. 0

Test Product	PQR No. RNC-PQR-004	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Test	Reduced Section Tension
Dimension	25 mm Thk.	Equipment's Capacity	200 TONS
Date Received	10 August 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	14 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	3181/12 TR1	N/A	25.00	20.07	501.750	N/A	N/A	N/A	216.530	431.550	N/A	OUT OF WELD
2	3181/12 TR2	N/A	24.84	20.41	506.984	N/A	N/A	N/A	221.041	435.992	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-004

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 16 / 08 / 12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 17 / 08 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



23859

TESTING
No. 0191

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-08-021

Revision No. 0

Test Product	PQR No. RNC-PQR-004	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Jig	Guided-Bend Roller Jig
Dimension	25 mm Thk.	Diameter of Mandrel	38.1 mm
Date Received	10 August 2012	Bend Angle	180 Degree.
Date Tested	14 August 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	3181/12 BS1	Transverse side bend	10.05x23.88x160.68 mm	No Open Discontinuity	N/A	-
2	3181/12 BS2	Transverse side bend	10.00x23.84x161.41 mm	Open Discontinuity	WELD ZONE	1.16, 0.19, 0.43 mm
3	3181/12 BS3	Transverse side bend	10.08x23.84x161.22 mm	Open Discontinuity	WELD ZONE	0.85 mm
4	3181/12 BS4	Transverse side bend	10.00x23.82x161.33 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-004

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date/...../.....

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date/...../.....

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

12748

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnrm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-08-034

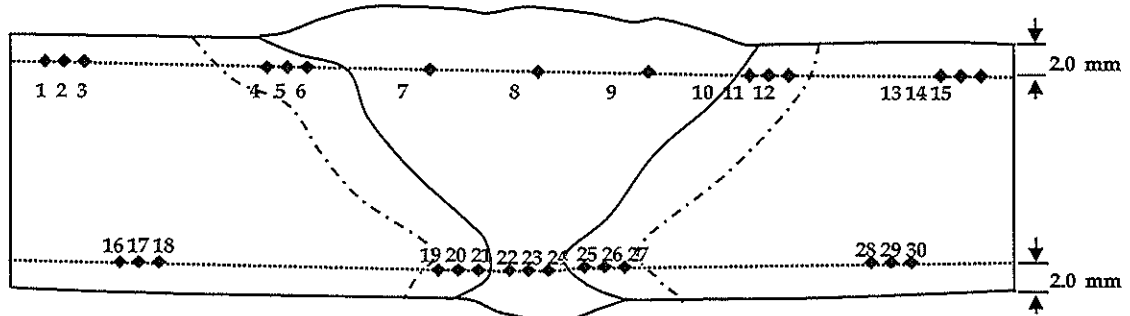
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-004	Material Characterization	Plate
Test No.	3181/12 HN	Test Method	ASTM E92-03
Project Name	N/A	Standard of Test	AWS D1.1/D1.1M : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	JIS G3101 SS400	Equipment	Mitutoyo, HV-115
Dimension	25 mm Thk.	Indentor Size	Diamond, pyramid 136 deg.
Welding Process/Position	SMAW/4G	Test Load	HV 10 Kgf.
Type of Joint	Butt joint	Total Force Dwell Time	15 s
Date Received	10 August 2012	Cal. Block	No. 62137 No. A62072
Date Tested	14 August 2012	Ambient Temperature	25.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	123.6	16	BASE	129.9	*****		
2	BASE	121.7	17	BASE	130.1			
3	BASE	121.7	18	BASE	129.4			
4	HAZ	147.7	19	HAZ	148.5			
5	HAZ	148.8	20	HAZ	157.6			
6	HAZ	150.8	21	HAZ	176.4			
7	WELD	161.9	22	WELD	194.7			
8	WELD	178.4	23	WELD	199.4			
9	WELD	163.7	24	WELD	199.0			
10	HAZ	163.7	25	HAZ	175.5			
11	HAZ	161.7	26	HAZ	161.7			
12	HAZ	152.4	27	HAZ	152.2			
13	BASE	128.6	28	BASE	132.9			
14	BASE	127.4	29	BASE	132.6			
15	BASE	126.5	30	BASE	130.9			

Additional details: Filler Metal : E7016(SMAW)

Reported by :

(Mr.Keerati Sa-ngoan)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 15/08/12

Notes

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LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Macrostructure Analysis Report

Page 1 of 1

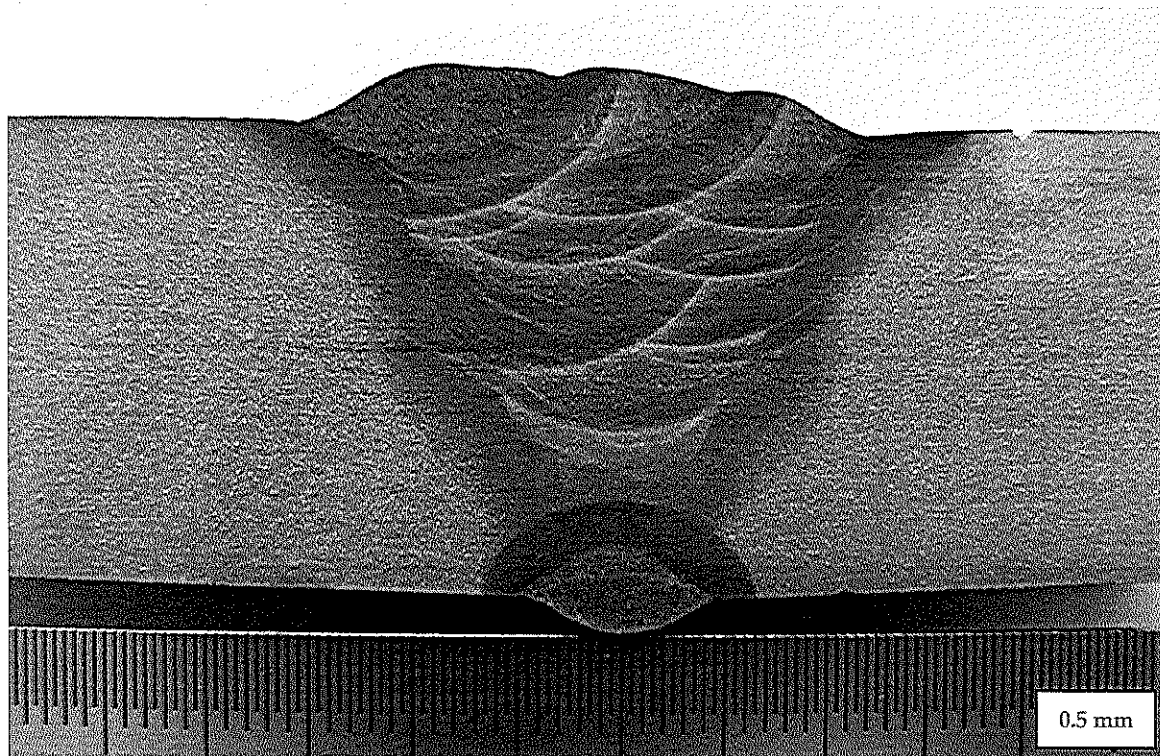
Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-08-025

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-004	Material Characterization	Plate
Test No.	3181/12 MA	Test Method	ASME E340-00
Project Name	N/A	Standard of Test	AWS D1.1/D1.1M : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	JIS G3103 SS400	Temperature	28 °C
Dimension	25 mm Thk.	Time of Etching	10 minutes
Welding Process/Position	SMAW/4G	Surface Preparation	Polishing with sand paper
Type of Joint	Butt joint		240, 400, 600, 800, 1000
Date Received	10 August 2012	Date Tested	14 August 2012

**Additional details :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Filler Metal : E7016(SMAW)

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 15/08/12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 15/08/12

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



3864

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-08-087

Revision No. 0

Test Product	PQR No. RNC-PQR-004	Material Characterization	Plate
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{e1}
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Test	Charpy V-Notch Impact
Dimension	25 mm Thk.	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	10 August 2012	Equipment 's Capacity	542 Joules
Date Tested	15 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	3181/12-1	-20°C	10.017	10.013	54.96	156.97	156.89	Weld Metal	-
2	3181/12-2	-20°C	10.016	10.015	54.96	157.22			
3	3181/12-3	-20°C	10.019	10.013	54.94	156.48			
4	3181/12-1	-20°C	10.019	10.014	54.91	325.03	283.73	Fusion Line+1mm	-
5	3181/12-2	-20°C	10.017	10.014	54.90	199.33			
6	3181/12-3	-20°C	10.000	10.013	54.89	326.82			
7	3181/12-1	-20°C	10.017	10.013	54.92	10.75	11.39	Fusion Line+5mm	-
8	3181/12-2	-20°C	10.017	10.013	54.93	10.65			
9	3181/12-3	-20°C	10.001	10.015	54.91	12.76			

Additional details : WPS No. RNC-WPS-004

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date/...../.....

Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date/...../.....

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LPN PLATE MILL

Tax sent by : 18-07-12 00:23 Pg: 1

FAX NO. : 20 Jul. 2012 4:47AM F1

Commodity : HOT ROLLED STEEL PLATES

Specification : REFER TO BS 4360-1990 GRADE 43A

INSPECTION CERTIFICATE



SAFLAVIRIYA PLATE MEL'POI

Office 28/1 Prapawit Bldg., 4th Fl., Surasak Rd., Silom, Bangkok, Bangkok,
Thailand 10500 Tel.: (662) 630-0257-66 Fax: (662) 630-0267

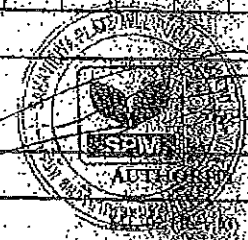
Factory 160 Moo 14, Sukhumvit Rd., Bangpakong, Chachoengsao,
Thailand 24130 Tel: (038)832-056-60 Fax: (038)531-635

ISSUE DATE: 24/05/2012

[illegible]

WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN SATISFACTORY TESTED

LABORATORY ENGINEER



20/7/54

KOBELCO

INSPECTION CERTIFICATE

PURCHASER

Date of Issue 11/Aug/2011

Certificate No. TF11-035

Date of Inspection 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION							
LB-52	3.2 x 350	12110	AWS A5.1 E7016							
Chemical Composition (%)										
Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97
Tensile Test of Weld Metal					Impact Test of Weld Metal					
Yield Strength at 0.2% offset	Tensile Strength		Elongation		Test Temp.	Absorbed Energy		Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009		
465 MPa	554 MPa		33 %		-30 °C	Avg.	246 J			
Welding Conditions					Postweld Heat Treatment			REMARKS		
Type of Current	AC									
Welding Current	120 A									
Welding Voltage	24 V									

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.



THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

